

Wearable Alert Device: Wiring Diagram

University of Oklahoma
School of Electrical and Computer Engineering
Spring 2026 Semester
Senior Capstone Project

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Adafruit Industries Lithium Ion Battery Pack – 3.7V 4400mAh
This battery can be switched out for a higher voltage battery if desired.

3.7V 4400mAh

LIPO Charger – Rechargeable component for the 3.7V Battery
This component is used to recharge the Lithium Battery Pack

LIPO_Charger1

The GPS will always be running and collecting data.
It collects longitude, latitude, and altitude of
the current location

GPS Tracking Module
Gravity_GNSS_Receiver_Module

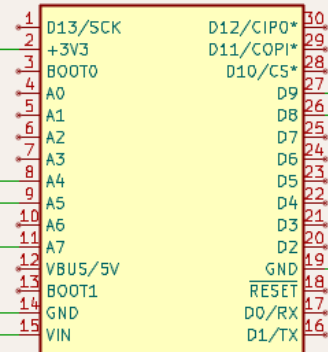


Microphone Detection
MEMS_Microphone



The analog microphone is constantly on and detecting.
If the reading is high enough for a gunshot detection,
it will send a signal.
If it is not high enough to be a gunshot, it will
not send a signal and continue to run like normal.

Arduino_Nano_ESP



Software Testing

R1 1kΩ D1 LED

LED turns on if a signal is sent.
This means if a Gunshot is detected
or if the emergency button is pressed.

Emergency Button

Push_Switch

This button can be pressed at anytime.
If the button is pressed and it is a valid
press, it will send a signal.

GND